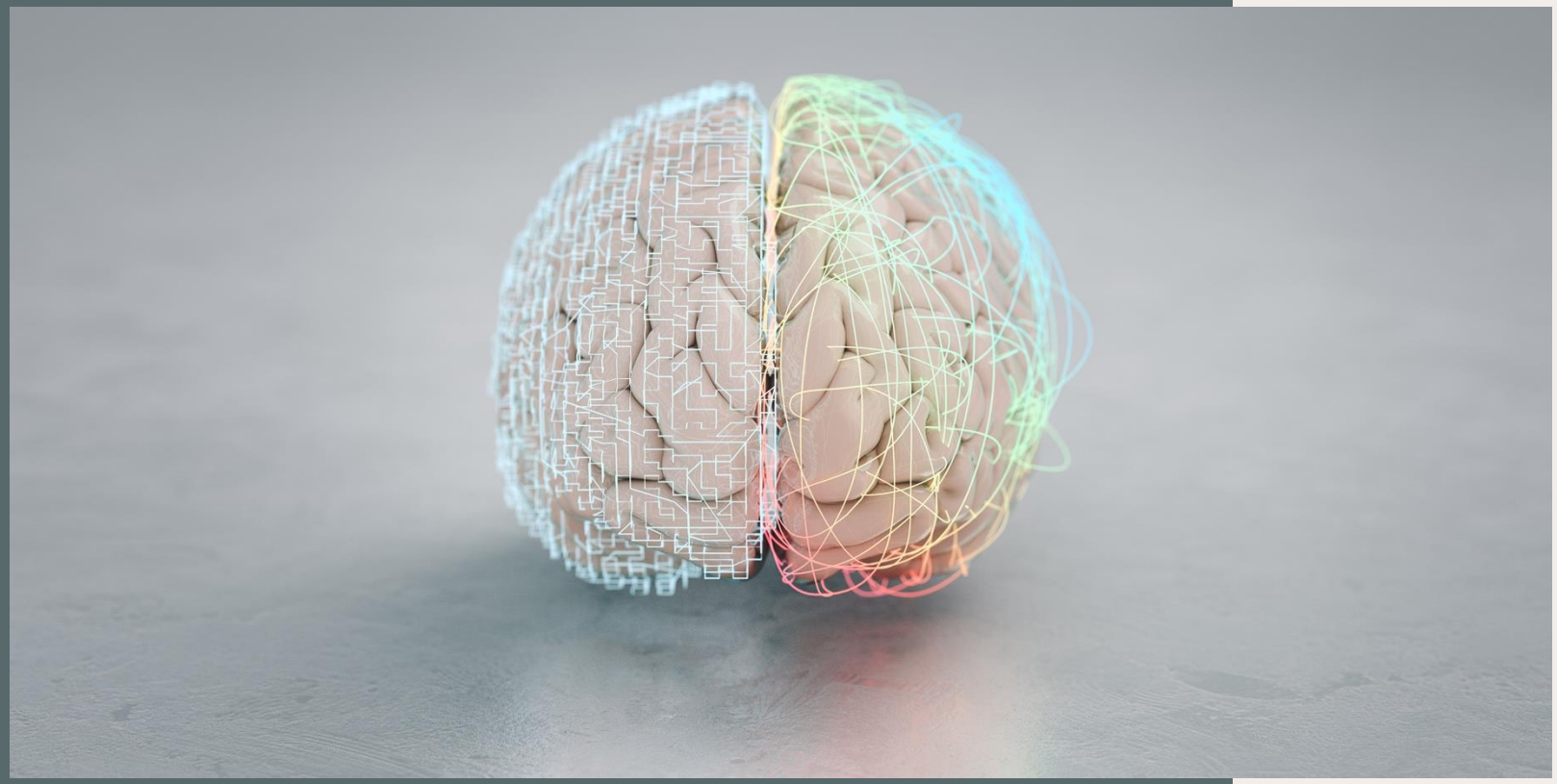


Characterizing and Improving the Reliability of Brain Connectivity extracted with Magnetic Resonance Imaging

Céline Provins

16/10/2026

CC-BY 4.0



Dr. Céline Provins

Doctorat en neuroimagerie à l'hôpital universitaire de Lausanne sous la supervision de Prof. Patric Hagmann et Dr. Oscar Esteban

Contribué à fMRIPrep et MRIQC

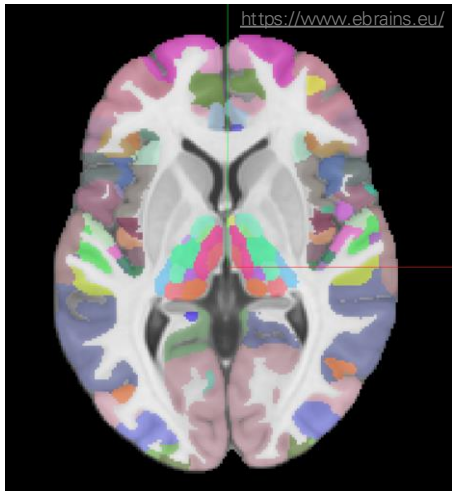
Data Scientist chez Greenteg

Would you build a house without verifying that the ground is solid enough?



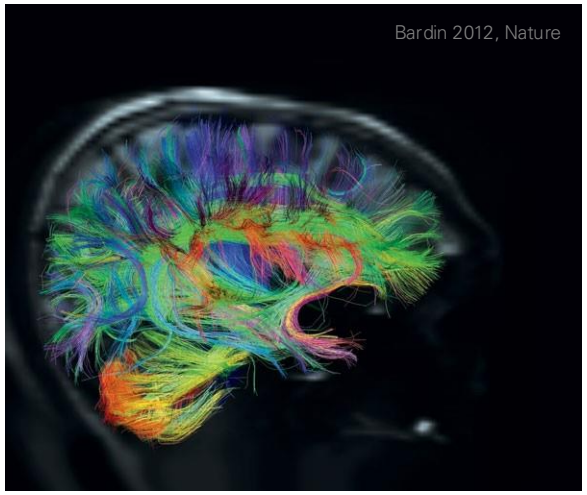
Structural connectivity (SC)

T1w



+

diffusion MRI

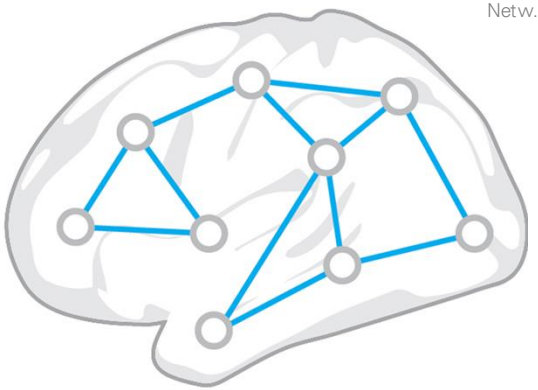


Quantify water diffusion

→

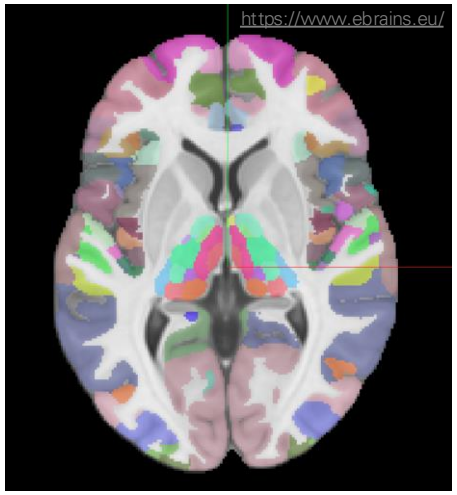
SC

[1] Faskowitz et al. 2022, Netw. Neurosci



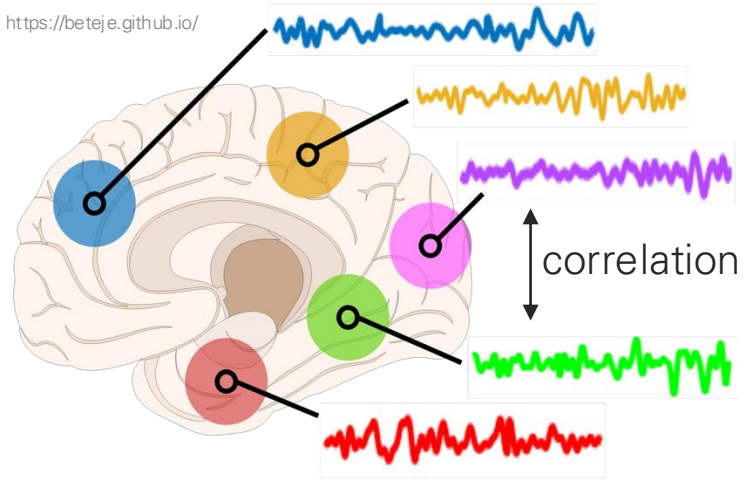
Functional connectivity (FC)

T1w



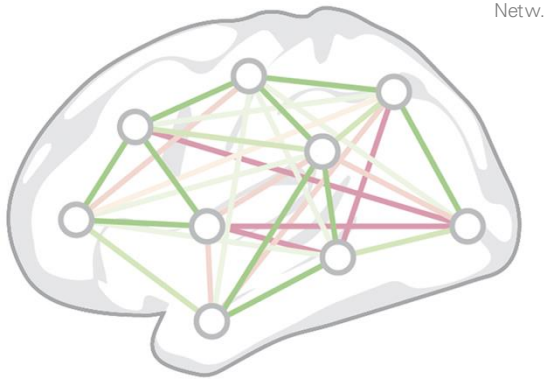
+

functional MRI



→

FC

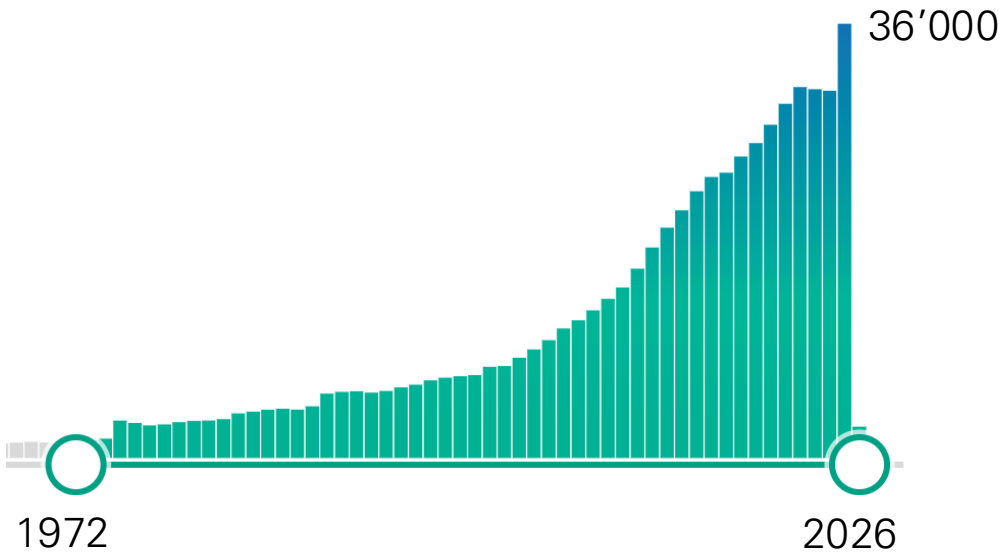


[1] Faskowitz et al. 2022, Netw. Neurosci

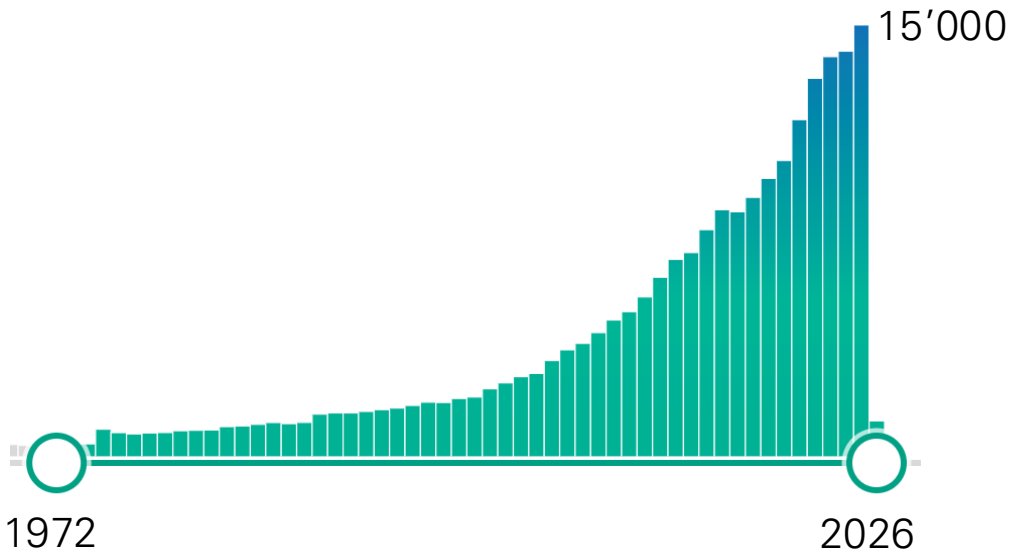
Functional and Structural Connectivity (FC/SC)

Searched term:[1]

Functional connectivity



Structural connectivity



[1]https://pubmed.ncbi.nlm.nih.gov/

Are brain connectivity estimates reliable?

- Reliable = produce consistent and reproducible results when repeated under similar conditions
- Whole-brain level [2,3,4,5,6] 
- Estimation of the individual connections value [5,7,8,9,10] 
- Connectivity matrices contain false positives and false negatives [8,9,10]

[2] Badhwar et al. 2020, NeuroImage
[3] O'Connor et al. 2017, GigaScience
[4] Cai et al. 2021, Magn. Reson. Med.
[5] Nakuci et al. 2023, Sci. Rep.
[6] Buchanan et al. 2014, NeuroImage
[7] Noble et al. 2019, NeuroImage
[8] Maier-Hein et al. 2017, Nat. Commun.
[9] Power et al. 2012, NeuroImage
[10] Zalesky et al. 2016, NeuroImage

How solid is the foundation on which are building our inferences about the brain ?

What are the main sources of variability that make FC and SC not reliable?

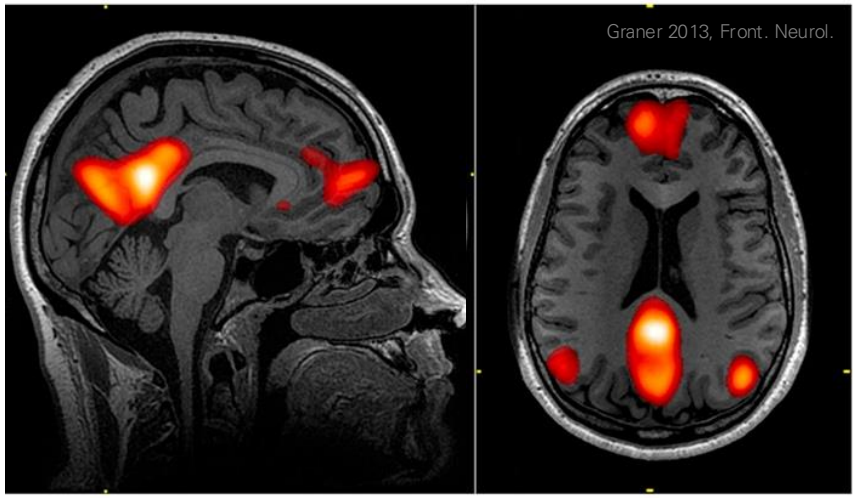
Highly indirect measures

diffusion MRI (dMRI)



Quantify water diffusion

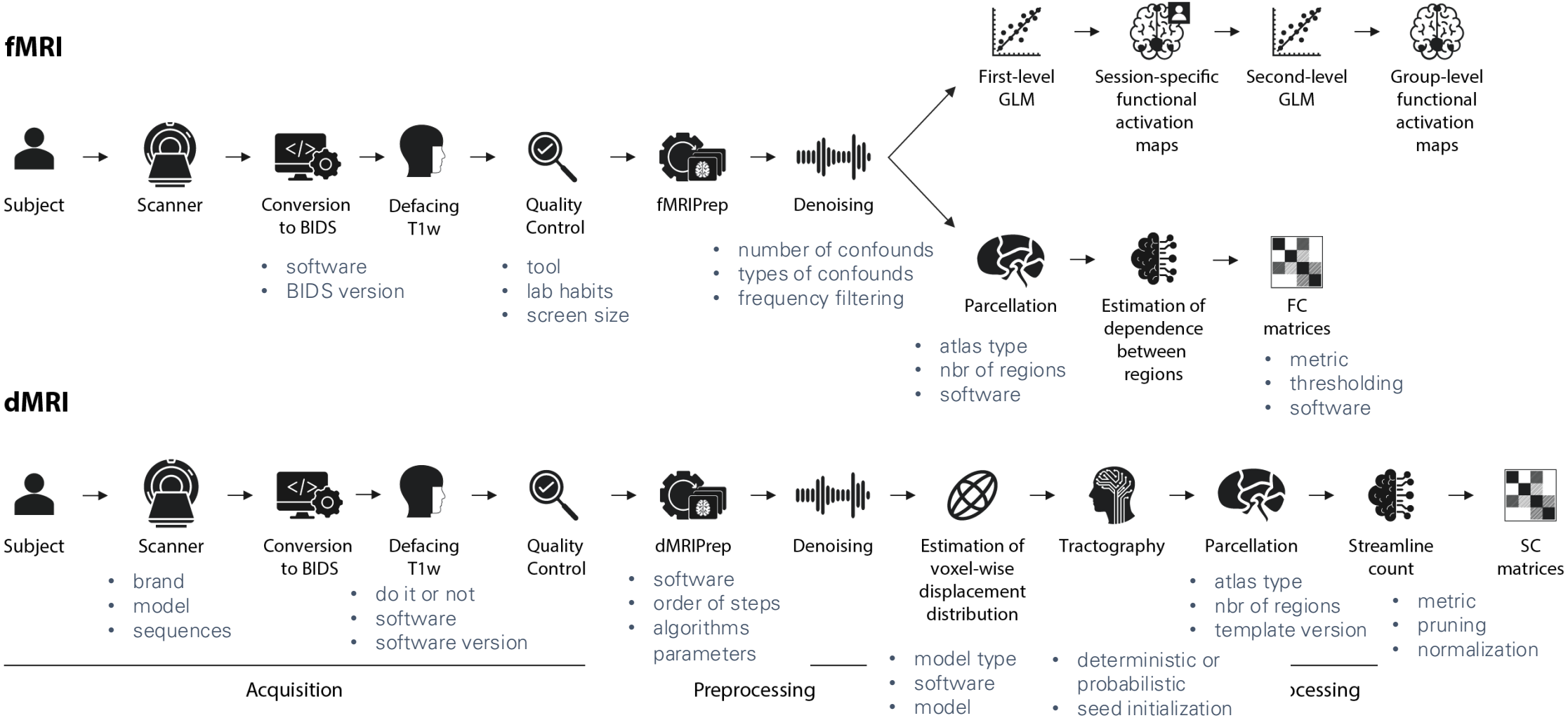
functional MRI (fMRI)



Derived from oxygenated blood concentration

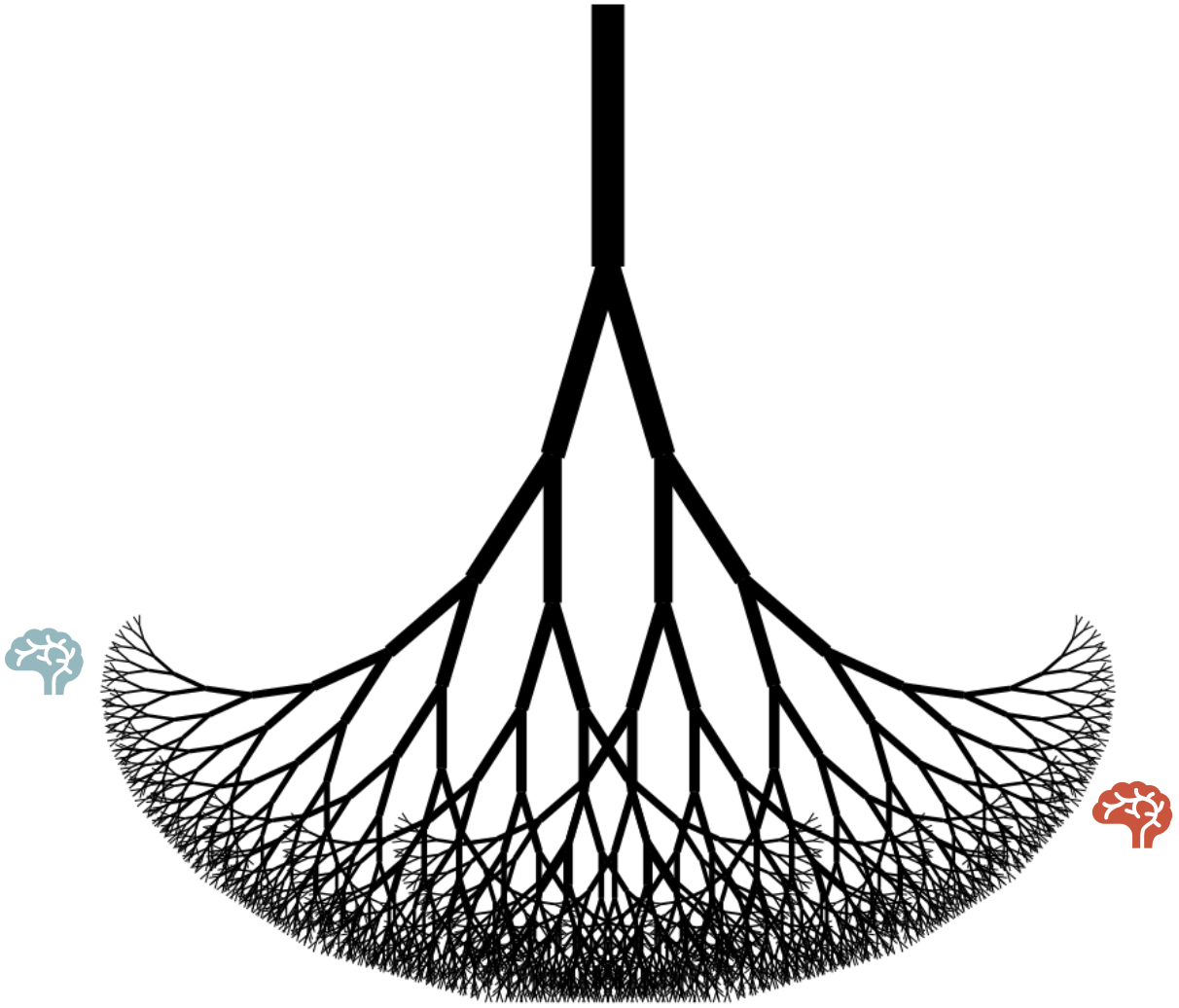
⇒ depends on the physiology, like heart rate and breathing

Many possible sources of variability due to long processing pipeline



Methodological variability

Subject

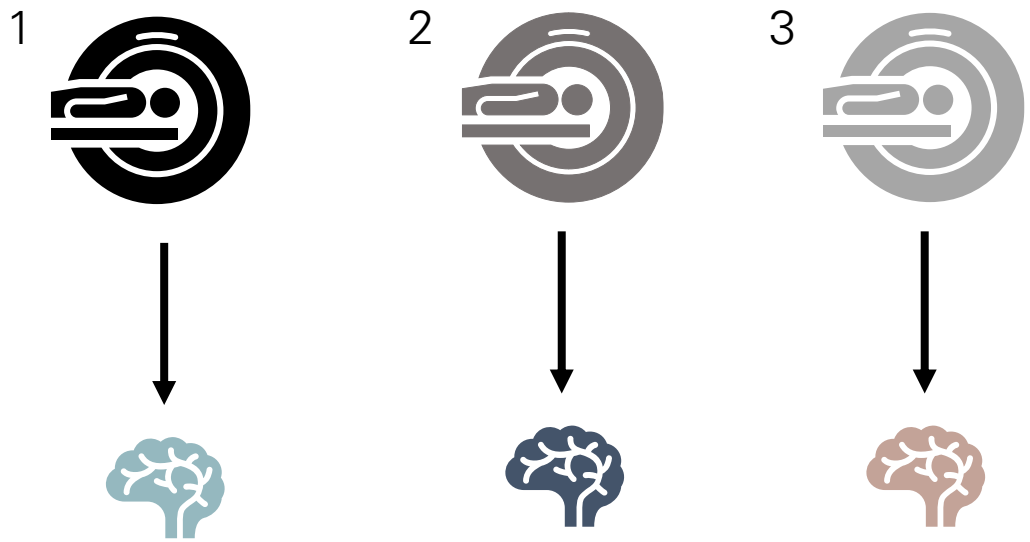


Brain connectivity estimates

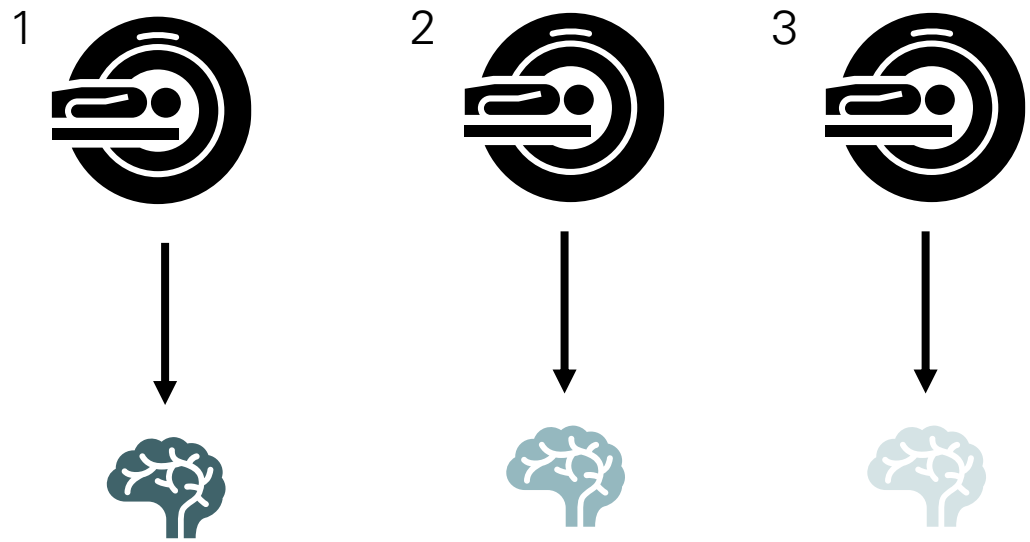
Par Claudio Rocchini — Travail personnel, CC BY-SA 3.0,
https://commons.wikimedia.org/w/index.php?curid=23529572

Scanner-level variability

Inter-scanner, e.g. different brand [3,11]



Intra-scanner, e.g. room temperature



[2] Badhwar et al. 2020, NeuroImage
[11] Magnotta et al. 2012, Brain Connect.

Subject-level variability

Inter-subject, e.g. different person



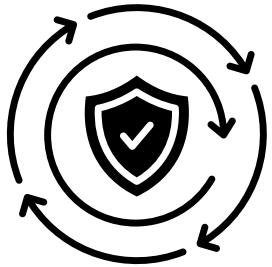
Intra-subject, e.g. consuming coffee or not [12]



[12] Poldrack et al. 2016, Nat. Commun.

Principles to improve neuroimaging reliability

Guiding principles to reliability



Redundancy



Standardization

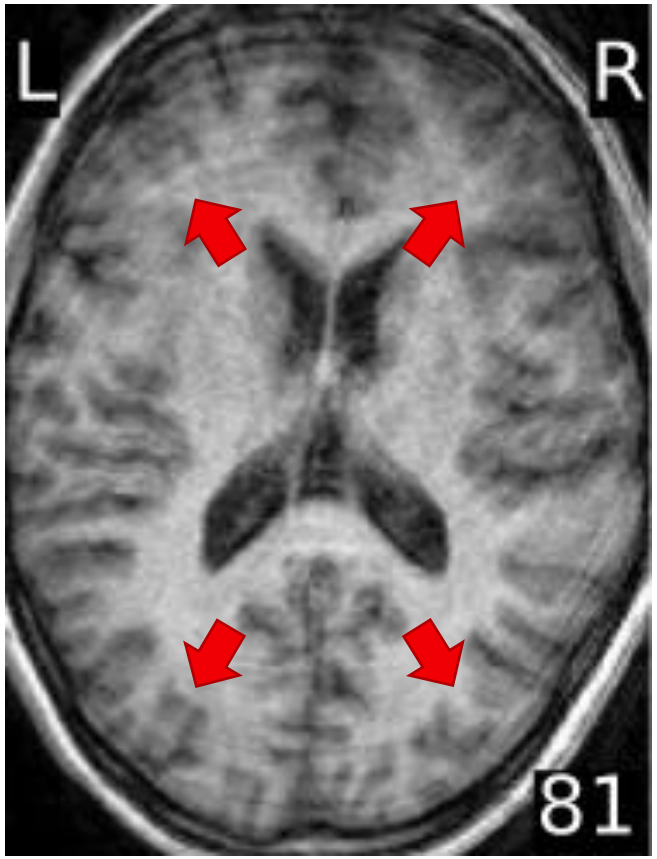
[13] Reading and Understanding More Multivariate Statistics

Registered Reports

- Defining the research plan before collecting or analyzing the data and submitting that research plan for peer review to a journal.
- Stage 1 acceptance in principle = journal commits to publishing the final manuscript regardless of the direction of the results, provided the pre-registered plan was rigorously followed.
- Stage 2 = final manuscript with results and discussion

Quality Control

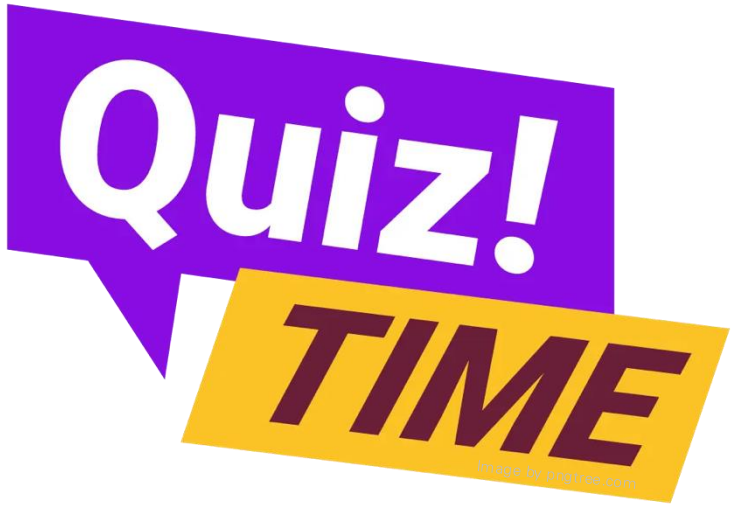
A



B



Participant moved



Which image is of better quality: A or B?

Quality Assurance and Quality Control (QA/QC)

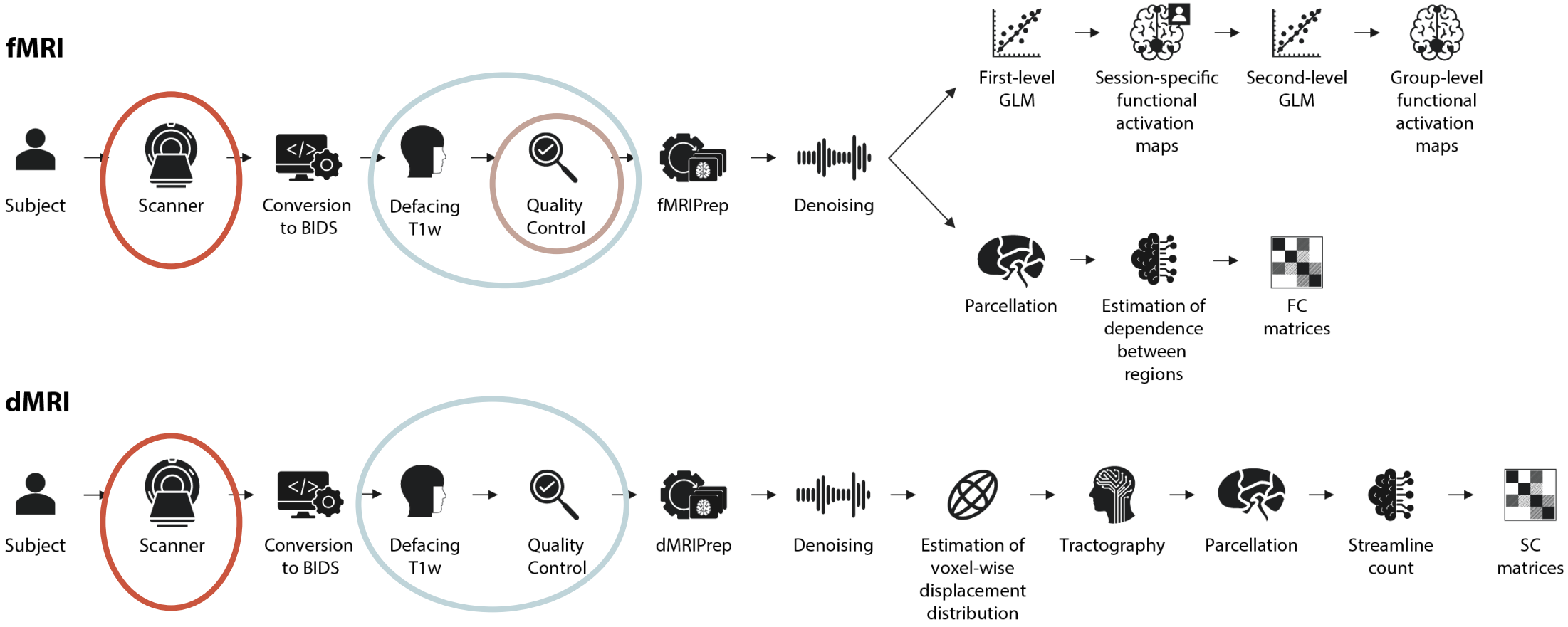
- Data of low-quality → erroneous conclusions [9, 14, 15]

- A precious way to understand the data and to gain confidence in its content [16, 17]

[9] Power et al. 2012, NeuroImage
[14] Alexander-Bloch et al. 2016, Hum. Brain Mapp.
[15] Ducharme et al.. 2016, NeuroImage
[16] Taylor et al. 2023, Front. Neurosci.
[17] Reynolds et al. 2023, Front. Neurosci.

Variability types tackled in this thesis

Variability tackled in this thesis

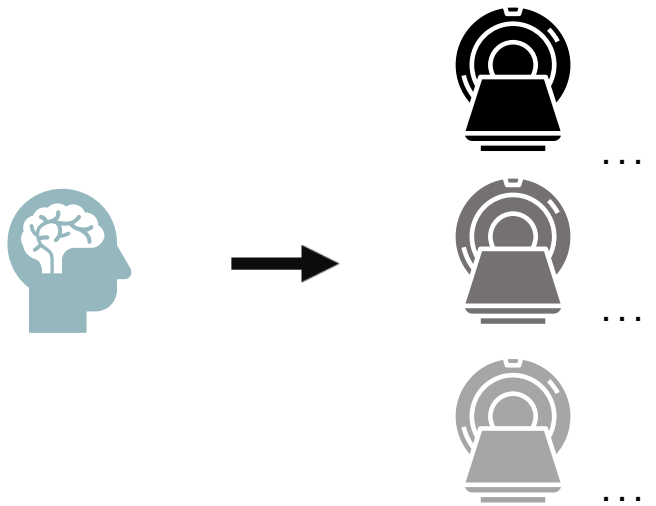


[18] Provins et al. 2023, Stage 1 Registered Report (RR) in Nature Methods
[19] Provins et al. 2025, Stage 2 RR in PLOS Biology
[20] Provins et al. 2023, Front. NeuroImaging

Isolate sources of scanner-level variability

Variability sources we want to isolate

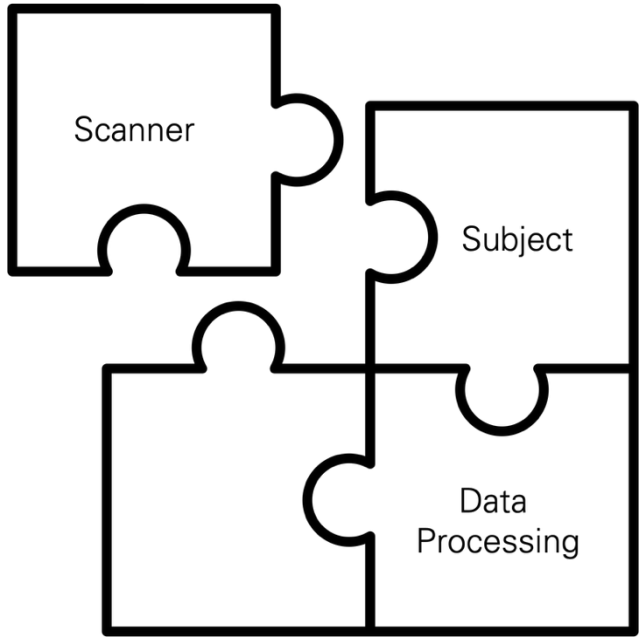
- Intra- and inter-scanner variability of → fMRI, dMRI
- Intra-subject variability related to physiology
    - to leverage physiological signal for denoising, need multi-echo fMRI



Open datasets

[12] Poldrack et al. 2016, Nat. Commun.
[21] Hank et al. 2014, Scientific Data
[22] O'Connor et al. 2017, GigaScience
[23] Pinho et al. 2018, Scientific Data
[24] Allen et al. 2022, Nat. Neurosci.
[25] Filevich et al. 2017,

Human Connectome Phantom (HCPh) dataset



Structural connectivity (SC)

~~Between-subject~~ [26, 27, 28]

Focus on a single subject

Within-subject

Between-scanner

Within-scanner

Methodological

Functional connectivity (FC)

~~Between-subject~~ [29, 30, 31]

Focus on a single subject

Within-subject

Between-scanner

Within-scanner

Methodological

[26] Buchanan et al. 2014, NeuroImage
[27] Cai et al. 2021, Magn. Reson. Med.
[28] Nakuci et al. 2023, Sci. Rep.
[29] Cabalo et al. 2024, bioRxiv
[30] Warrington et al. 2023, Imaging Neurosci
[31] Finn et al. 2015, Nat. Neurosci

Structural connectivity (SC)

~~Between-subject~~

Focus on a single subject

~~Within-subject~~

Acquisition in short time window

Between-scanner

Within-scanner

Methodological

Functional connectivity (FC)

~~Between-subject~~

Focus on a single subject

Within-subject

Between-scanner

Within-scanner

Methodological

Structural connectivity (SC)

~~Between-subject~~

Focus on a single subject

~~Within-subject~~

Acquisition in short time window

Between-scanner

Within-scanner

Methodological

Functional connectivity (FC)

~~Between-subject~~

Focus on a single subject

Within-subject

Between-scanner

Within-scanner

Methodological

Structural connectivity (SC)

~~Between-subject~~

Focus on a single subject

~~Within-subject~~

Acquisition in short time window

Between-scanner

Within-scanner

Methodological

Functional connectivity (FC)

~~Between-subject~~

Focus on a single subject

Between-scanner

Within-scanner  Within-subject

- Control subject cognition with a naturalistic paradigm [32]
- Minimize circadian effect [33]
- Measure physiological signals
- Measure confounds such as mood, sleep …

Methodological

[32] Finn 2021, Trends Cogn Sci
[33] Farahani et al. 2022, NeuroImage

Structural connectivity (SC)

~~Between-subject~~

Focus on a single subject

~~Within-subject~~

Acquisition in short time window

Between-scanner

Within-scanner

~~Methodological~~ [8, 34, 35]

Minimize it through standardization

Functional connectivity (FC)

~~Between-subject~~

Focus on a single subject

Between-scanner

Within-scanner  Within-subject

- Control subject cognition with a naturalistic paradigm [32]
- Minimize circadian effect [33]
- Measure physiological signals
- Measure confounds such as mood, sleep …

~~Methodological~~ [36, 37, 38, 39]

Minimize it through standardization

[8] Maier-Hein et al. 2017, Nat. Commun.
[34] Bhagwat et al. 2021, Gigasciences
[35] Thomas et al. 2014, PNAS
[36] Botvinik-Nezer et al. 2020, Nature
[37] Li et al. 2024, Nat. Commun.
[38] Luppi et al. 2024, Nat. Commun.
[39] Bowring et al. 2019, Hum. Brain Mapp.

Intra-scanner variability characterization

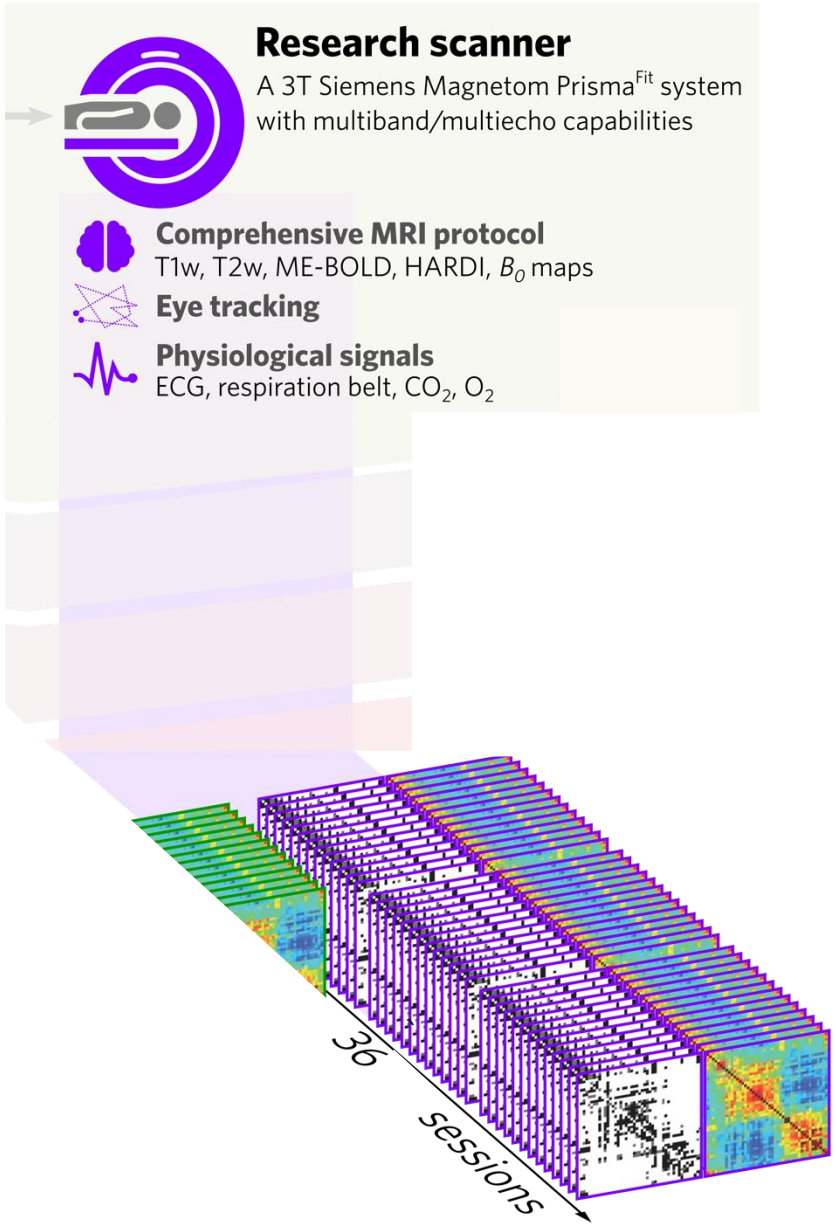
Human Phantom
One subject,
72 MRI sessions

Research scanner
A 3T Siemens Magnetom Prisma^{Fit} system
with multiband/multiecho capabilities

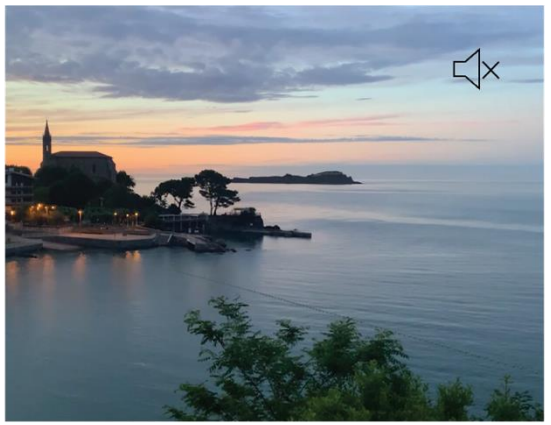
Comprehensive MRI protocol
T1w, T2w, ME-BOLD, HARDI, B_0 maps

Eye tracking

Physiological signals
ECG, respiration belt, CO_2, O_2

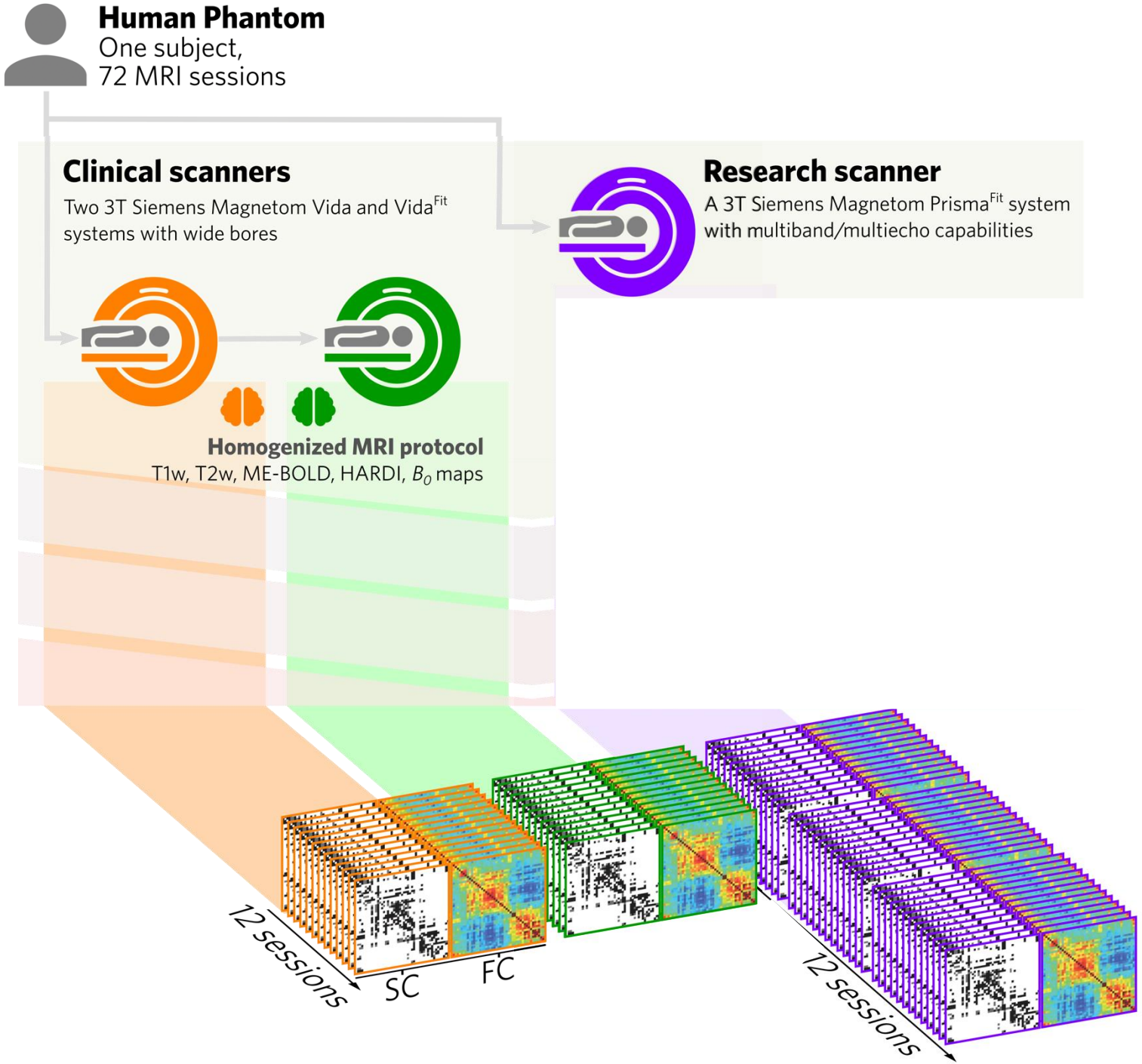


- Resting-state fMRI using naturalistic stimuli

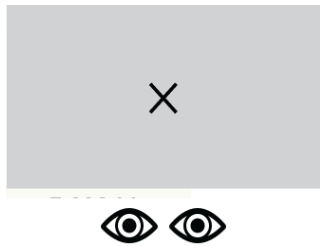


[18] Provins et al. 2023, Nat. Methods

Inter-scanner variability characterization



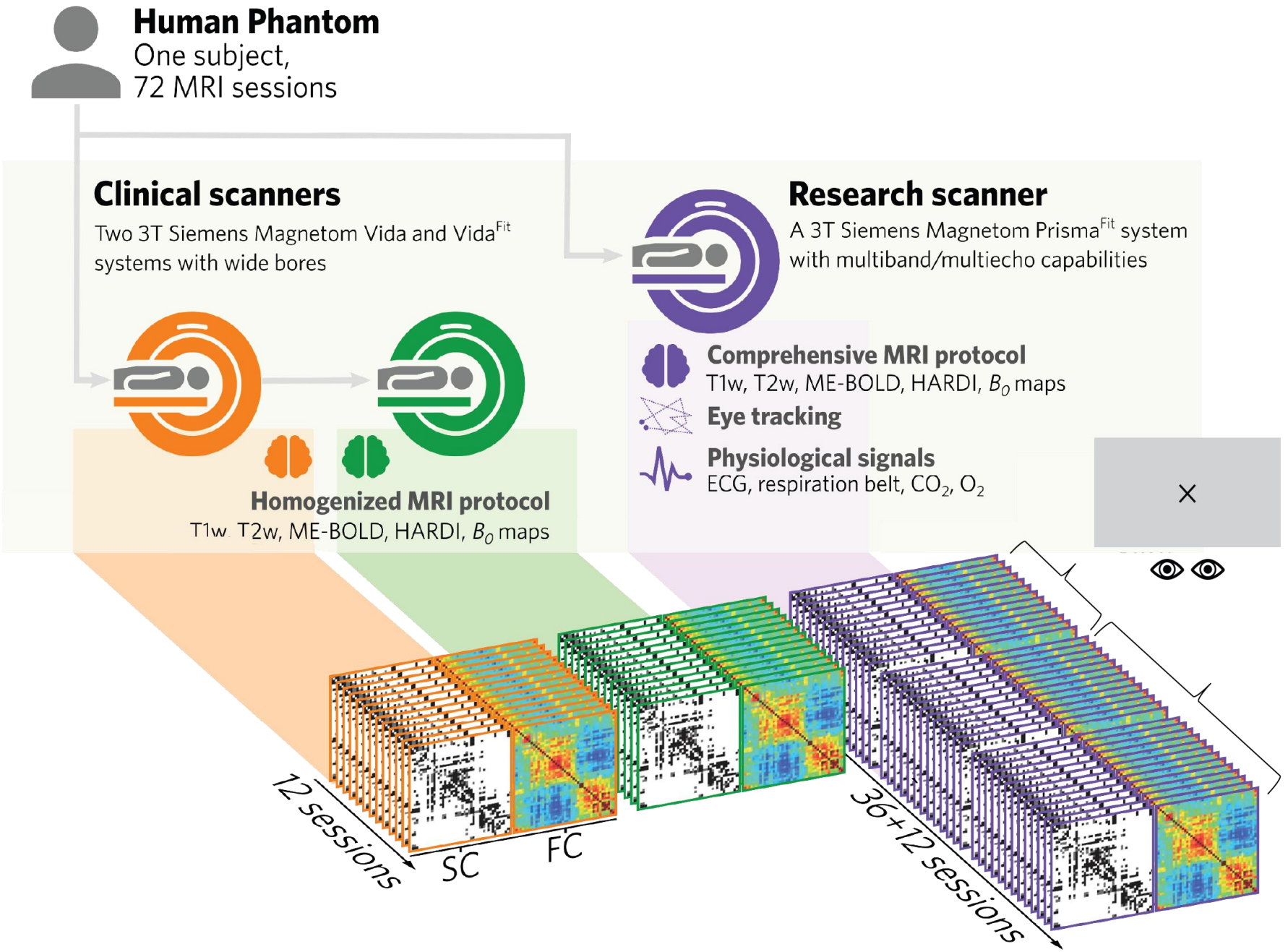
- Standard resting-state fMRI eyes-open

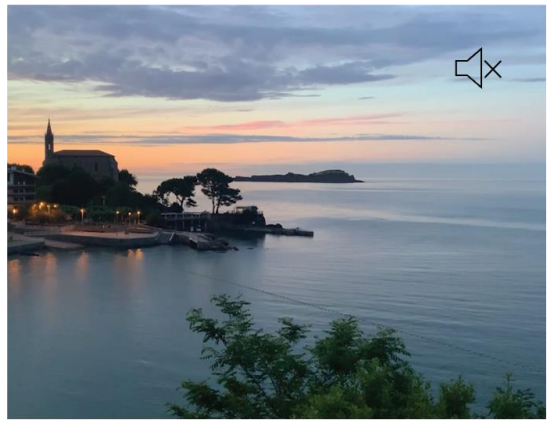


[18] Provins et al. 2023, Nat. Methods

Ranking of variability sources

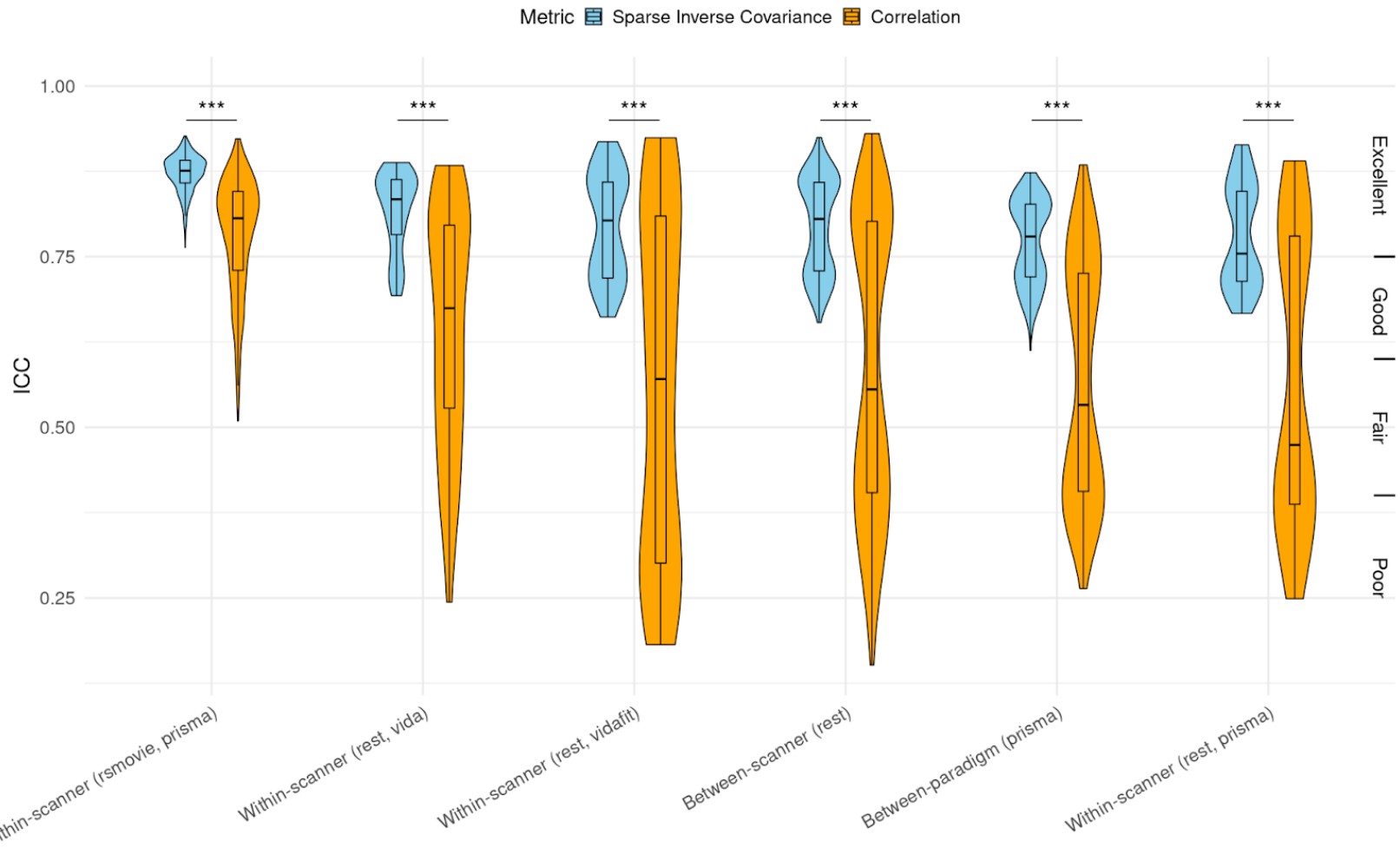
- Isolated and quantified 6 variability types in a single framework





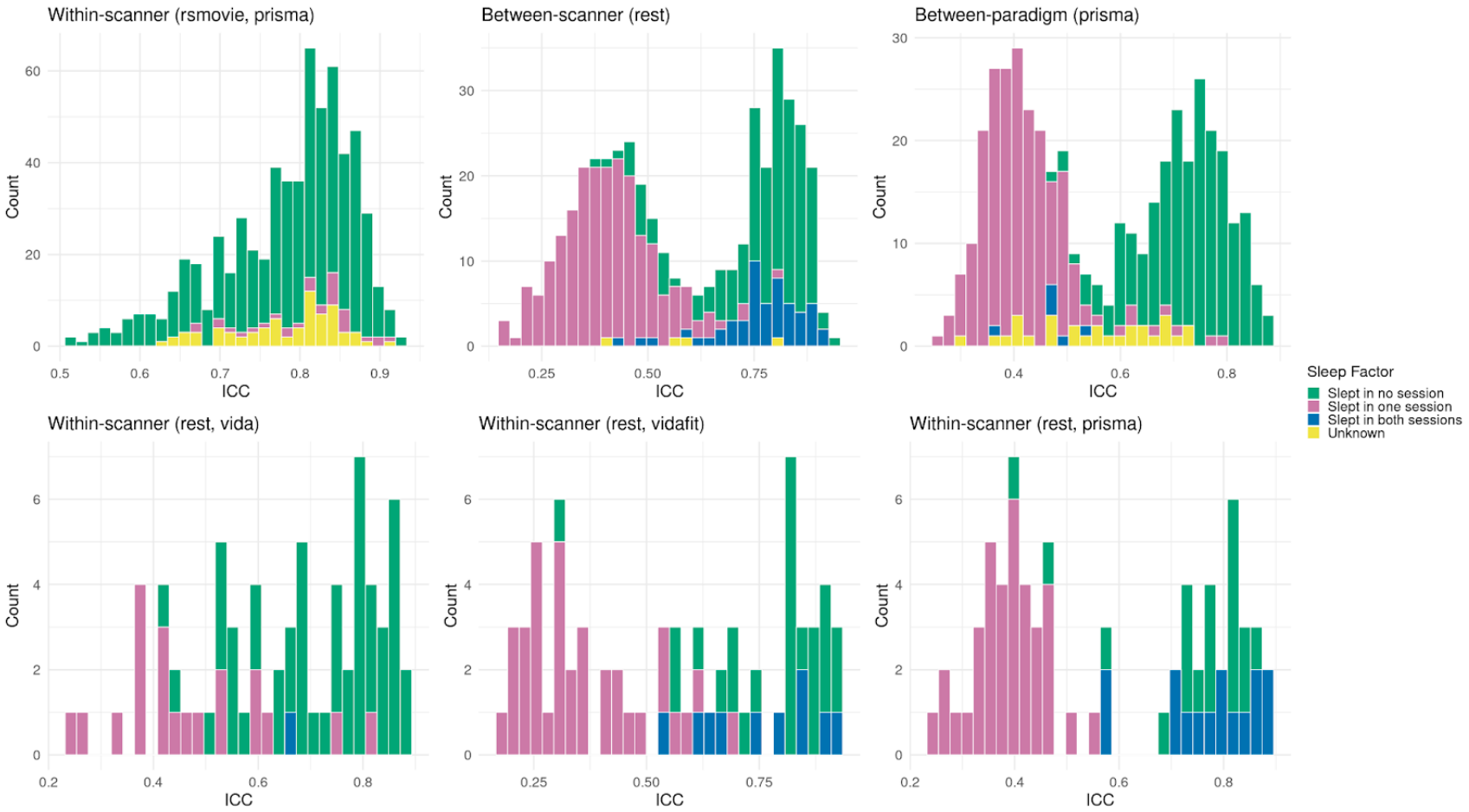
[18] Provins et al. 2023, Nat. Methods

We identified two concrete strategies to improve the reliability of FC estimates



- Using sparse inverse covariance to estimate statistical dependence between timeseries resulted in more reliable FC estimates than correlation
  - Trade-off: loses sensitivity
- between-scanner < within-scanner
- Most reliable = naturalistic movie

Naturalistic movie yield higher reliability by reducing sleepiness



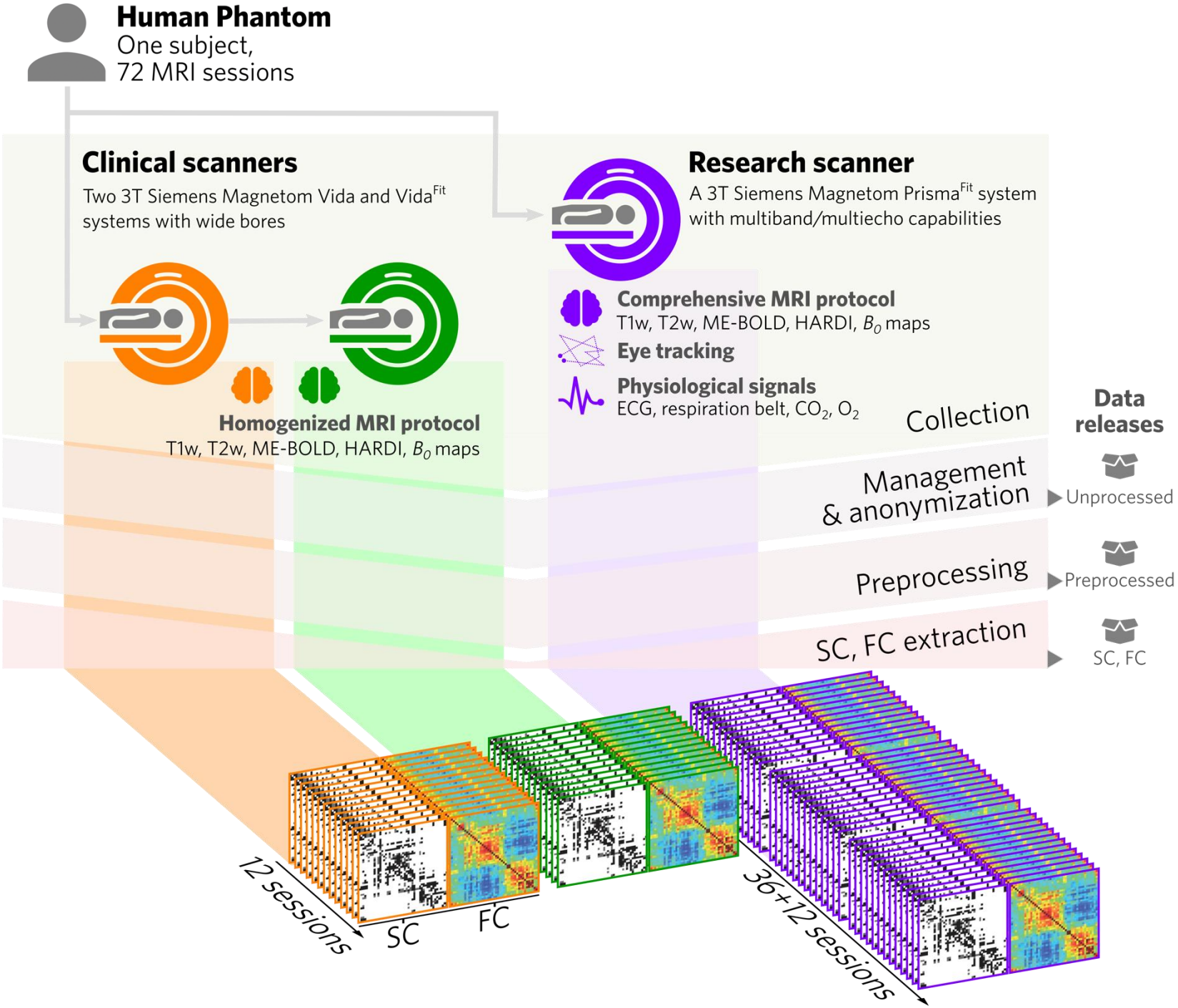
- We did not find evidence of reduced variability across movie repetition following hypothetical habituation[41]

- Participant fell asleep

30%	3%
Standard resting-state sessions	Naturalistic movie sessions

[40] Provins et al., OHBM 2026, Bordeaux
[41] Provins et al., OHBM 2025, Brisbane

Rich and unique dataset

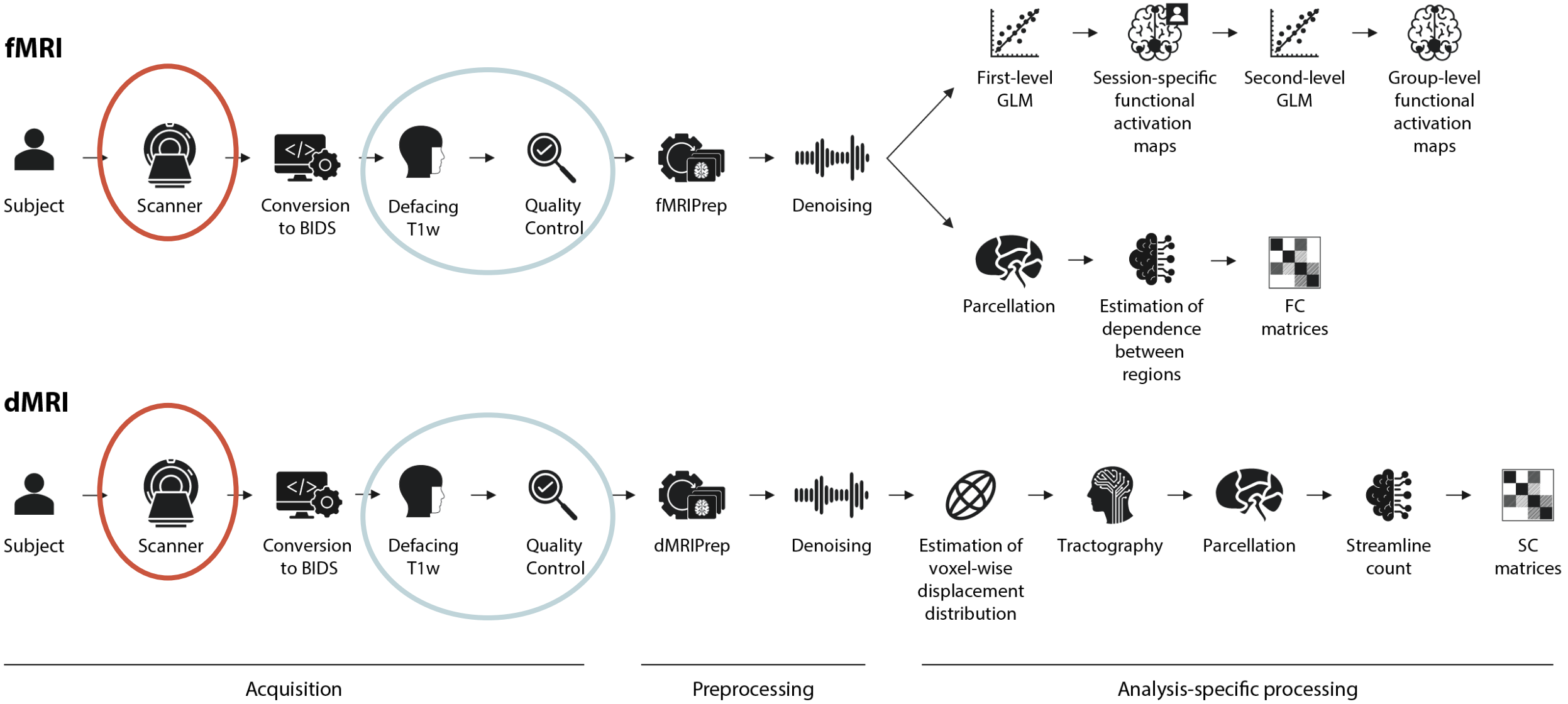


Many more interesting acquisition details which open up numerous research avenues

Especially around physiological signals

→ release data publicly to maximize its potential

[18] Provins et al. 2023, Nat. Methods

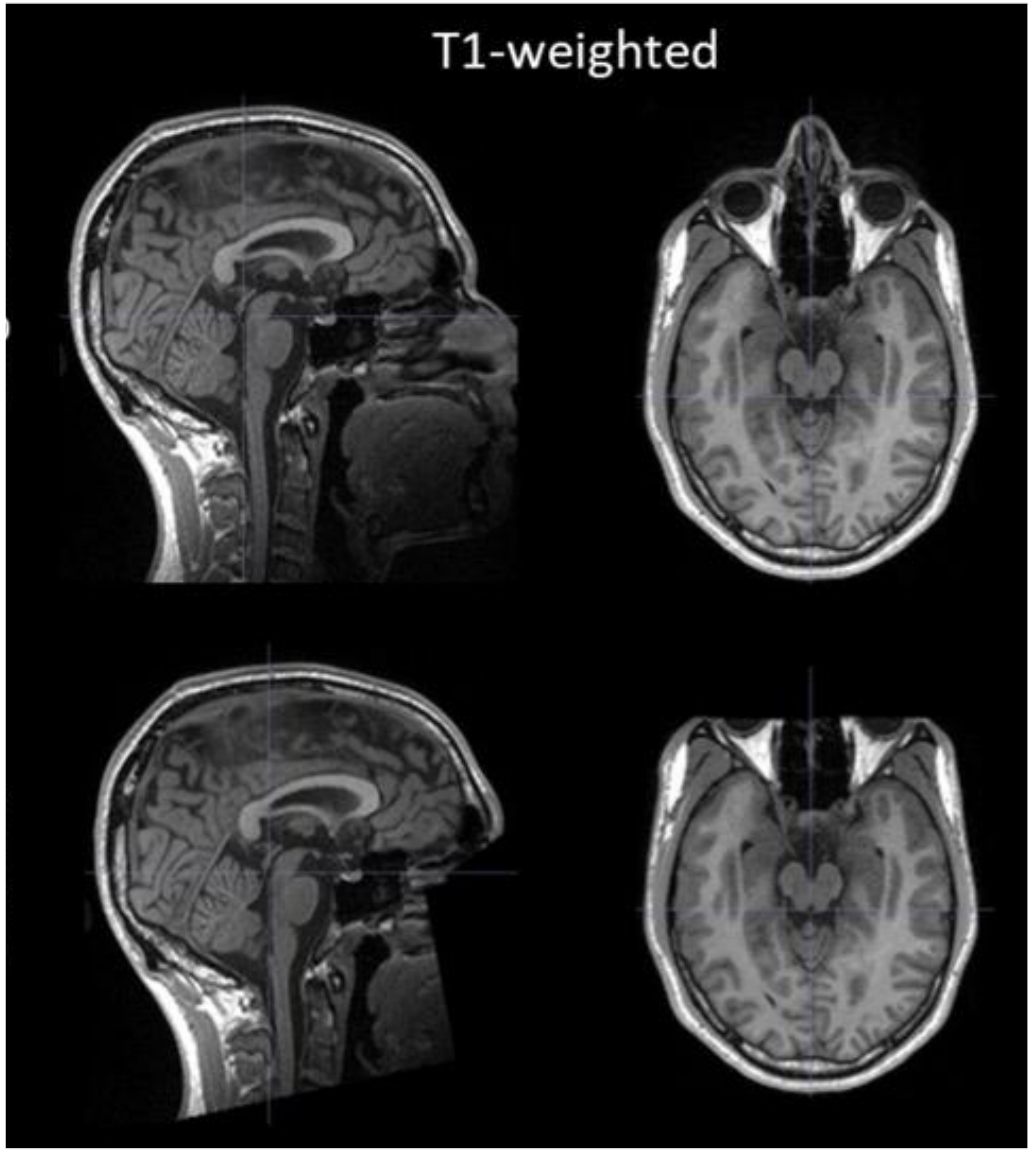

fMRI
Subject
Scanner
Conversion to BIDS
Defacing T1w
Quality Control
fMRIPrep
Denoising
First-level GLM
Session-specific functional activation maps
Second-level GLM
Group-level functional activation maps
Parcellation
Estimation of dependence between regions
FC matrices
dMRI
Subject
Scanner
Conversion to BIDS
Defacing T1w
Quality Control
dMRIPrep
Denoising
Estimation of voxel-wise displacement distribution
Tractography
Parcellation
Streamline count
SC matrices
Acquisition
Preprocessing
Analysis-specific processing





What is the difference between the images up and down?

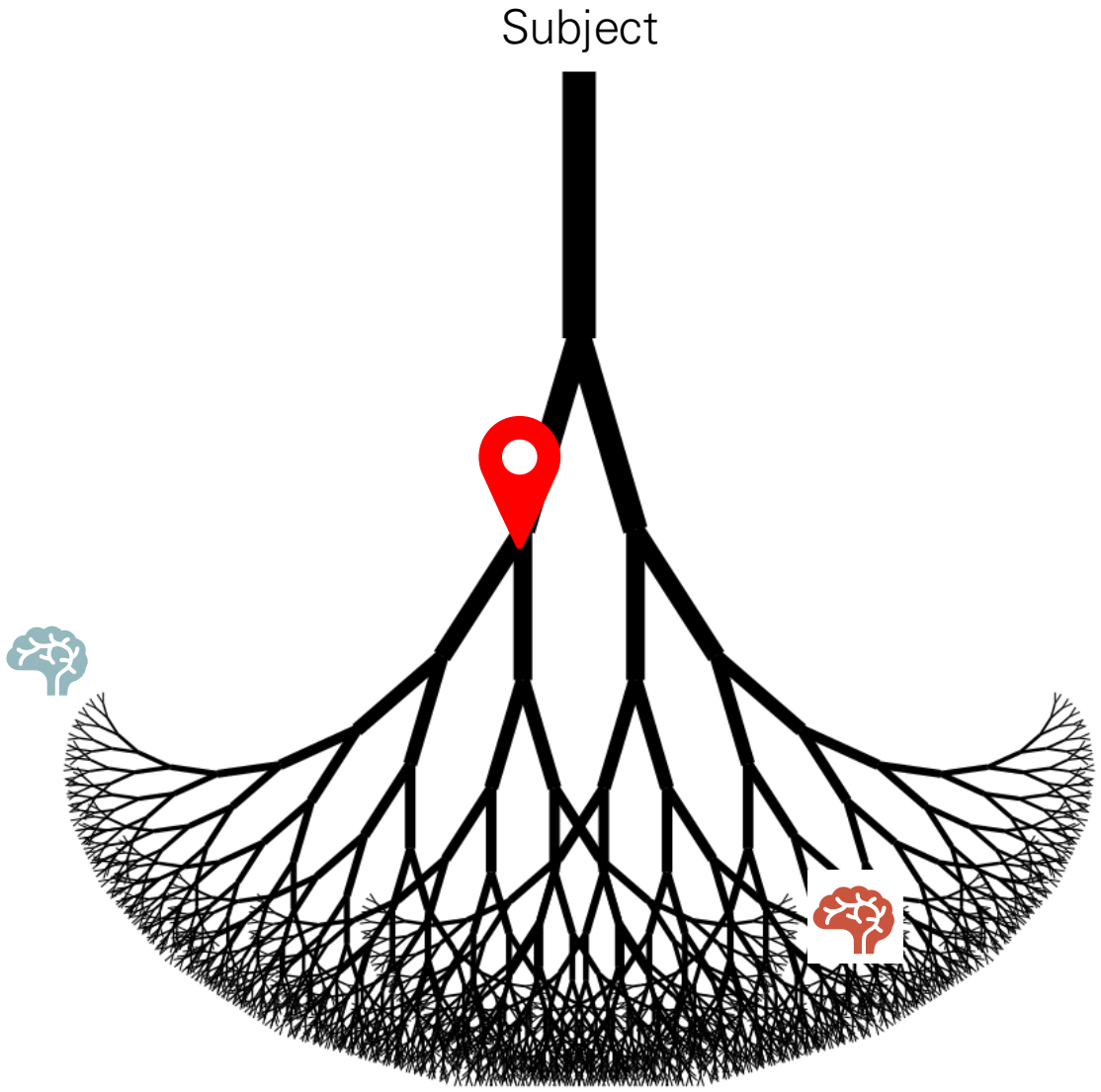
Why studying defacing influence on quality control ?



- Destructive step that affects image background
- Yet, background is very important for QC [20,42]
- QC is crucial for reliability
- Evidence that defacing has deleterious effects on downstream research workflow [43,44] and on automatic quality measurements [45]

[20] Provins et al. 2023, Front. Neuroimaging
[42] Provins et al. 2022, OHBM
[43] Schwarz et al. 2021, NeuroImage
[44] de Sitter et al. 2020, European Radiology
[45] Bhalerao et al. 2022, Neuroradiol J

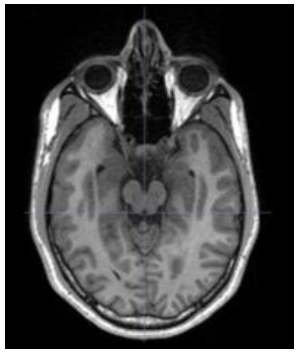
Substantial potential cascading effects

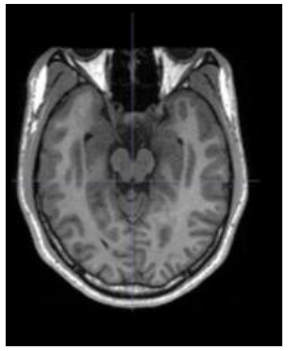


Brain connectivity estimates

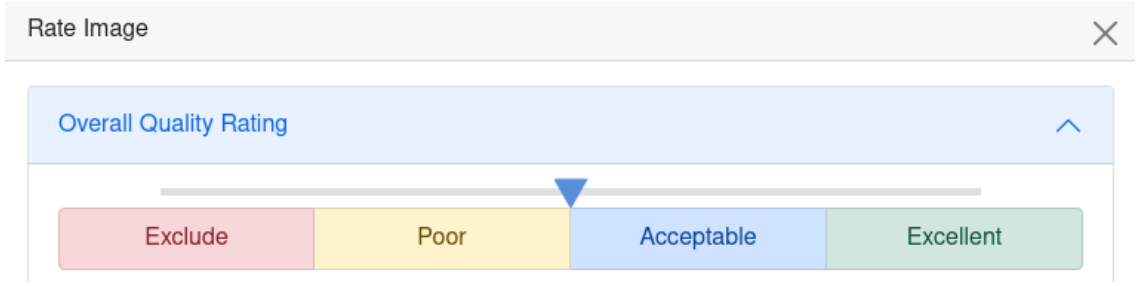
Par Claudio Rocchini — Travail personnel, CC BY-SA 3.0,
https://commons.wikimedia.org/w/index.php?curid=23529572

Data



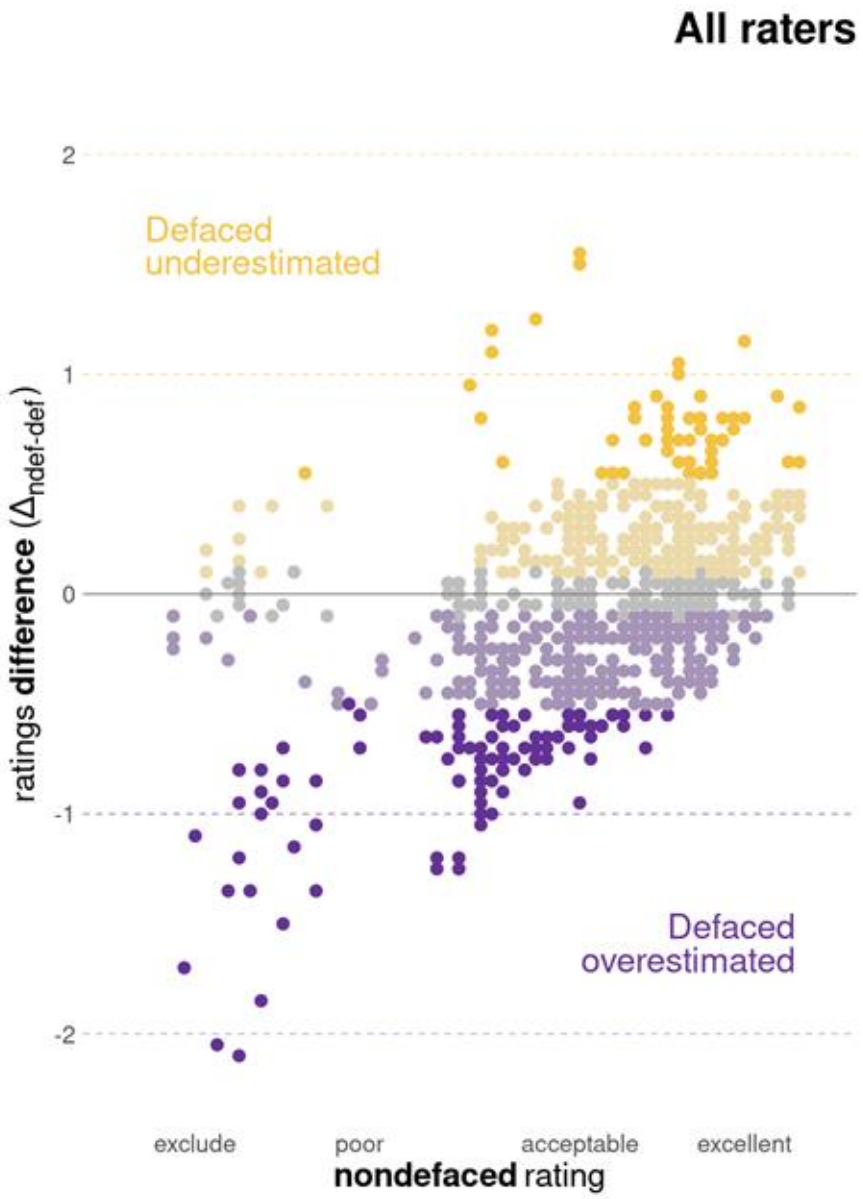


- 4 raters
- 185 images
- Rated using MRIQC in two conditions: nondefaced and defaced
- Total of 185 * 2 = 370 rated images



[46] Esteban et al. 2016, Plos One
[47] Hill et al. 2006, https://brain-development.org/ixi-dataset/

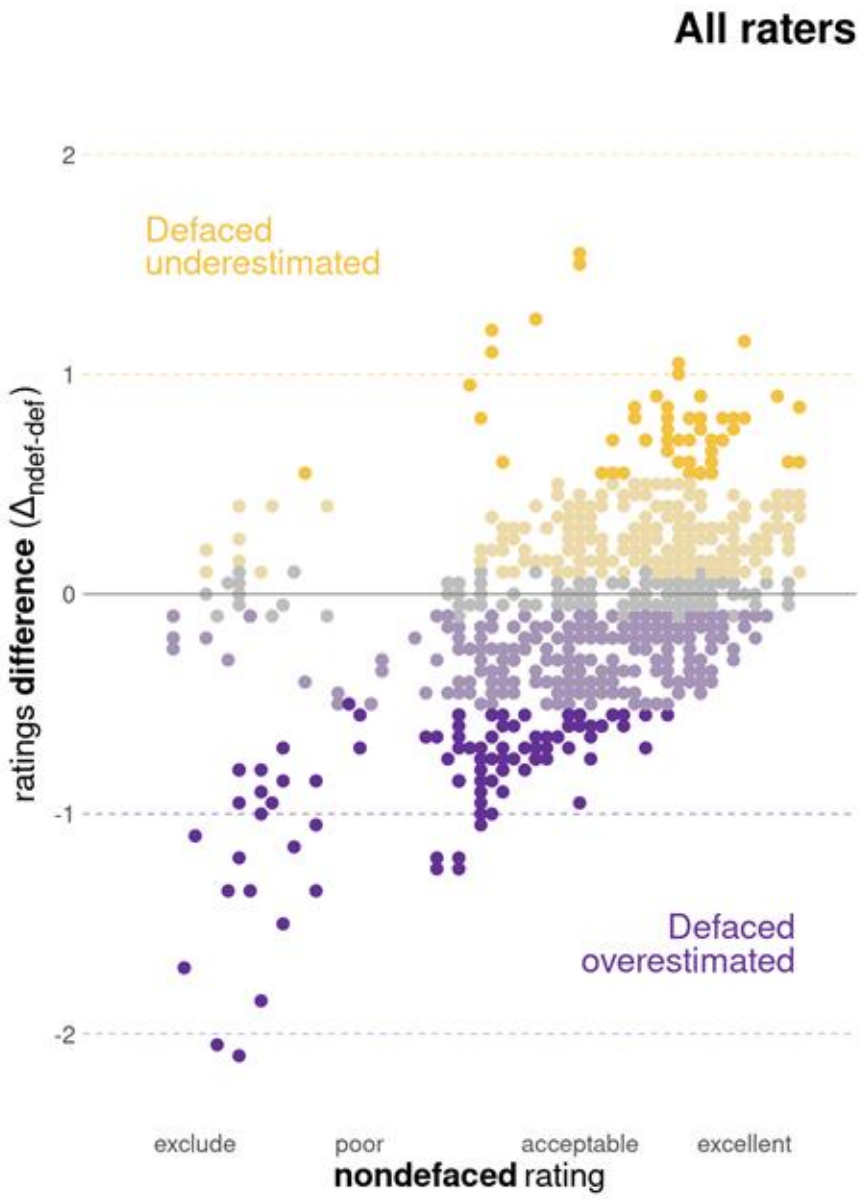
Bland-Altman plot to study reliability



- image rated in the nondefaced and defaced condition by one rater

[48] Altman and Bland 1983, J. R. Stat. Soc.

Defacing biased quality assessments by human raters



$n_{>0.5}$ = 268

max bias = 1.55

n_{stable} = 752

$n_{<0.5}$ = 440

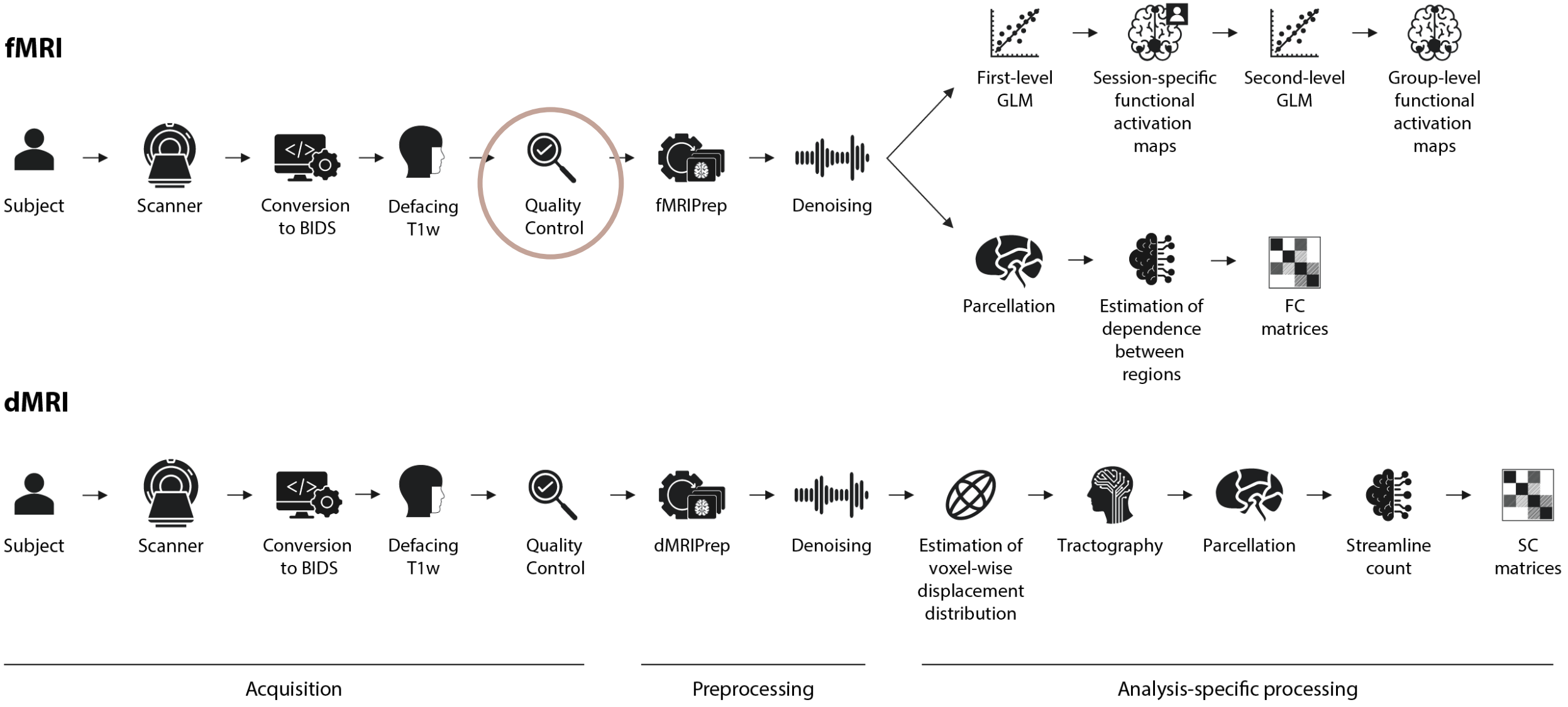
min bias = -2.1

Defacing removed critical information needed to evaluate image quality.

Impact

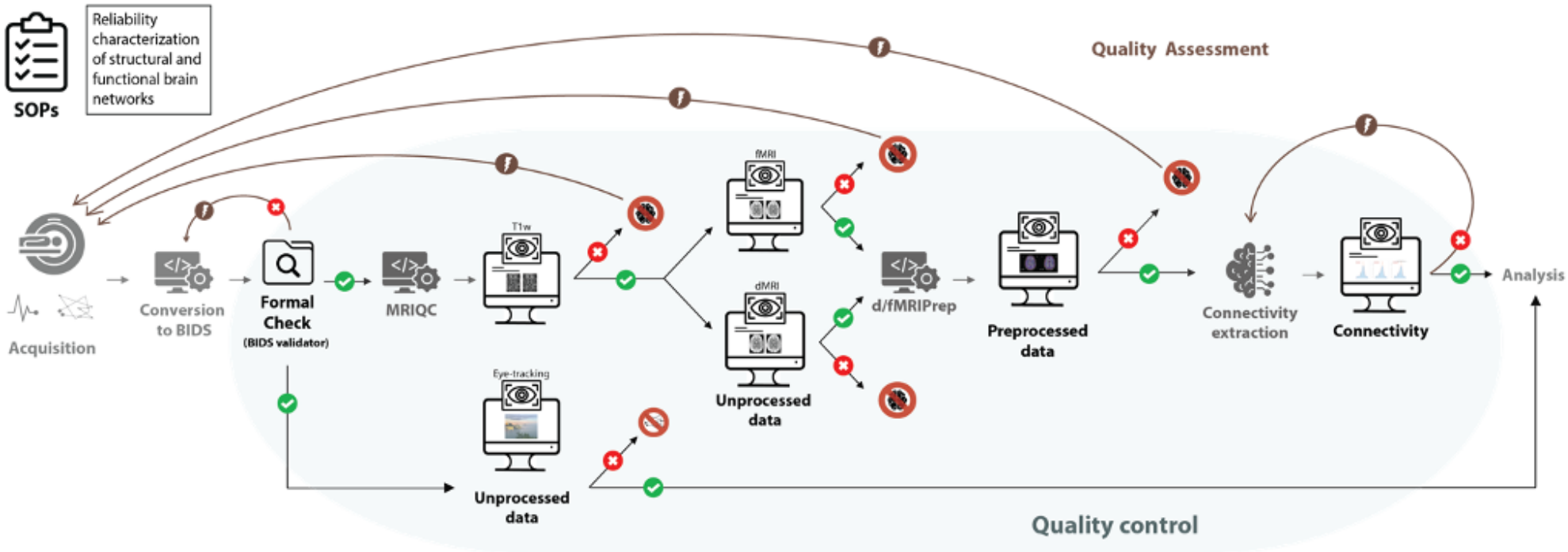
Defacing influenced human perception of quality

→ QC should be performed before defacing


fMRI
Subject
Scanner
Conversion to BIDS
Defacing T1w
Quality Control
fMRIPrep
Denoising
First-level GLM
Session-specific functional activation maps
Second-level GLM
Group-level functional activation maps
Parcellation
Estimation of dependence between regions
FC matrices
dMRI
Subject
Scanner
Conversion to BIDS
Defacing T1w
Quality Control
dMRIPrep
Denoising
Estimation of voxel-wise displacement distribution
Tractography
Parcellation
Streamline count
SC matrices
Acquisition
Preprocessing
Analysis-specific processing

Quality control protocol

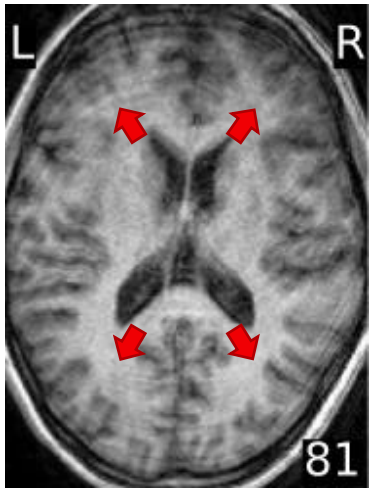
- Collaborative project
- Guide on how to read the visual reports generated by, among others, MRIQC
- 4 checkpoints



[20] Provins et al. 2023, Front. NeuroImaging

Precise exclusion criteria

Each checkpoint comes with a list of QC criteria to apply



QC of unprocessed T1w data based on the MRIQC visual report	J) Artifactual structures in the background K) Susceptibility distortion artifacts KA) Signal drop-out KB) Brain distortions L) Aliasing ghost M) Wrap-around that overlaps with the brain N) Data formatting issues O) Motion-related and Gibbs ringing P) Extreme intensity non-uniformity Q) Eye spillover

The order of the criteria is arbitrary.

[20] Provins et al. 2023, Front. NeuroImaging

QC decisions depend on the research context

QC criteria must reflect the intended analysis, available data and corrections and acceptable trade-offs

fMRI

98%

Excluded (178/181)

Frequent signal drop-out

No fieldmaps available for correction

→ Strict exclusion

T1w

31%

Flagged (56/181)

Potentially problematic artifacts

But used for normalization and co-registration

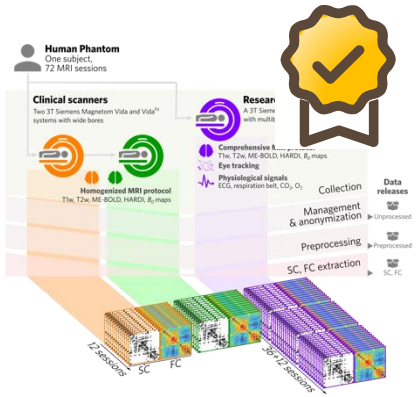
→ Inspect, don't automatically exclude

The same artifact does not always require the same decision

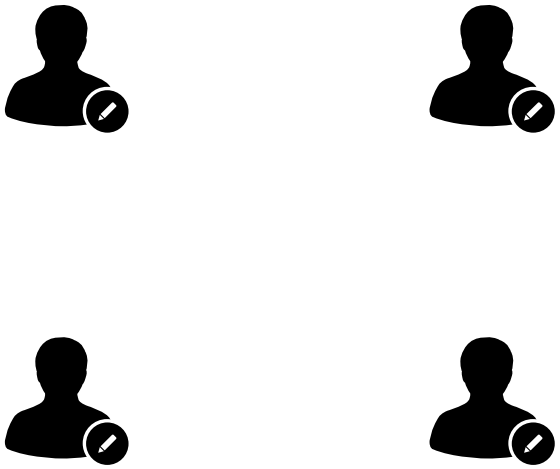
[20] Provins et al. 2023, Front. NeuroImaging

This protocol contributed to the other two projects

Ensuring the quality
of our HCPh dataset

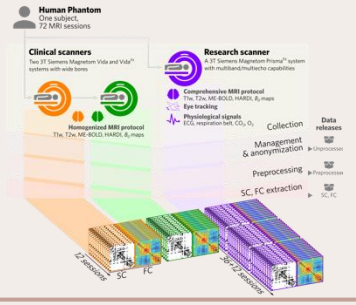


Reduce the variability
between the raters in the
defacing project

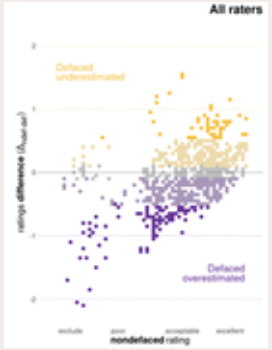


Conclusion

We created a dataset tailored to isolate inter- and intra-scanner variability in both fMRI and dMRI



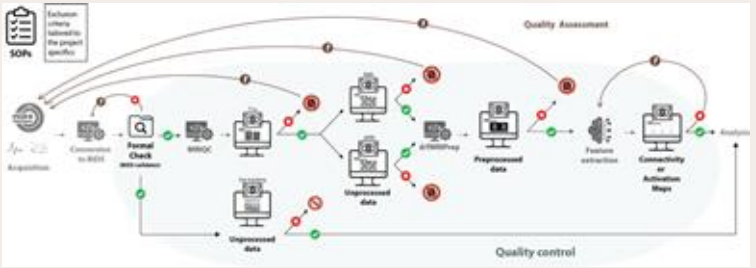
Actions early in the pipeline may have significant downstream effects.



Identified 4 concrete strategies to improve reliability of FC studies

1. Perform QC before defacing
2. Using same scanner when possible
3. Acquire with naturalistic movie
4. Use sparse inverse covariance

I developed a comprehensive QA/QC protocol



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Acknowledgments





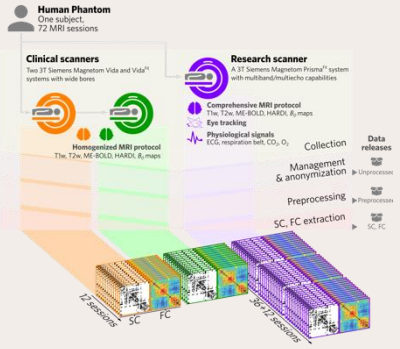


Thank you to all collaborators
and
a special thanks to Dr. Esteban!

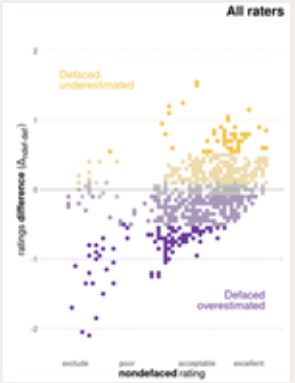
FNSNF Fonds national suisse de la recherche scientifique
CHUV
Unil UNIL | Université de Lausanne
LEMANIC NEURO SCIENCE

Conclusion

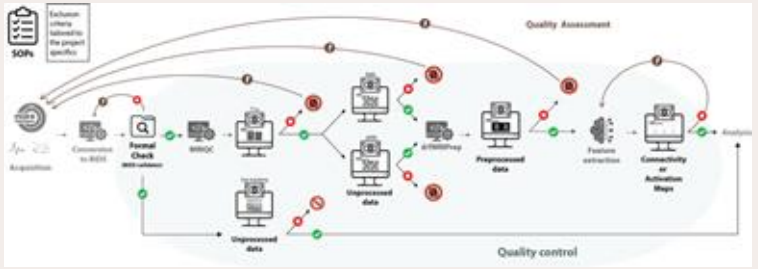
HCPh Dataset Stage 1 Registered Report



Defacing Stage 2 Registered Report



QA/QC protocol



Thank you for your attention!

Happy to answer your questions